

Wumpus World in Situation Calculus — Q2 Notes

The three things every action needs

For each action A you must give:

1. A **possibility axiom** $\text{Poss}(A, s) \iff \dots$ (when can A be performed?)
2. A **successor-state axiom** for every fluent it can change, of the form

$$F(\vec{x}, \text{RESULT}(a, s)) \iff \underbrace{\Phi^+(\vec{x}, a, s)}_{a \text{ made it true}} \vee \left(F(\vec{x}, s) \wedge \neg \underbrace{\Phi^-(\vec{x}, a, s)}_{a \text{ made it false}} \right)$$

This single schema combines effect axioms *and* frame axioms, solving the frame problem.

If a fluent has no successor-state axiom, you have implicitly claimed nothing changes it — often wrong. Write one axiom *per fluent*, not per action.

Ontology for Wumpus World

Constants & functions

- Locations are pairs $[x, y]$. The agent is the constant `AGENT`.
- Orientations: 0, 90, 180, 270 (degrees).
- `LOCATIONTOWARD` $([x, y], d)$: the adjacent cell in direction d .
- `LOCATIONAHEAD` $(s) = \text{LOCATIONTOWARD}(\text{LOC}(s), \text{ORI}(s))$ where `LOC` (s) , `ORI` (s) are the unique location/orientation of `AGENT` in s .
- `ADJACENT` (p_1, p_2) , `WALL` (p) — static map predicates (not fluents).

Fluents

- `AT` $(\text{AGENT}, [x, y], s)$ — agent's location
- `ORIENTATION` (AGENT, d, s) — direction faced
- `HAVEARROW` (s)
- `WUMPUSALIVE` (s)

Helper

$$\begin{aligned} \text{FACINGWUMPUS}(s) \iff & \exists [x, y]. \text{AT}(\text{WUMPUS}, [x, y], s) \\ & \wedge \text{INLINEOFFIRE}(\text{AGENT}, [x, y], s) \end{aligned}$$

where `INLINEOFFIRE` says the wumpus shares the row/column ahead of the agent's current orientation. Treat `WUMPUS`'s location as constant once placed.

(a) Shoot

Possibility. Can only shoot if you have an arrow:

$$\text{POSS}(\text{SHOOT}, s) \iff \text{HAVEARROW}(s)$$

Successor-state for HaveArrow. Any SHOOT consumes the arrow; nothing else affects it:

$$\text{HAVEARROW}(\text{RESULT}(a, s)) \iff \text{HAVEARROW}(s) \wedge a \neq \text{SHOOT}$$

Successor-state for WumpusAlive. The wumpus dies iff currently alive and the agent shoots while facing it:

$$\begin{aligned} \text{WUMPUSALIVE}(\text{RESULT}(a, s)) \iff & \text{WUMPUSALIVE}(s) \\ & \wedge \neg(a = \text{SHOOT} \wedge \text{FACINGWUMPUS}(s)) \end{aligned}$$

(b) Move (Forward)

Possibility. You can move forward iff the square ahead exists and is not a wall:

$$\text{POSS}(\text{FORWARD}, s) \iff \neg \text{WALL}(\text{LOCATIONAHEAD}(s))$$

Successor-state for At. A FORWARD action puts you on the square ahead; any other action leaves the location alone:

$$\begin{aligned} \text{AT}(\text{AGENT}, p, \text{RESULT}(a, s)) \iff & (a = \text{FORWARD} \wedge p = \text{LOCATIONAHEAD}(s)) \\ & \vee (a \neq \text{FORWARD} \wedge \text{AT}(\text{AGENT}, p, s)) \end{aligned}$$

The second disjunct is the frame part: every non-FORWARD action preserves location.

(c) Orient (TurnLeft / TurnRight)

Define rotation functions

$$\text{LEFTTURN}(d) = (d + 90) \bmod 360, \quad \text{RIGHTTURN}(d) = (d - 90) \bmod 360.$$

Possibility. Turning is always allowed:

$$\text{POSS}(\text{TURNLEFT}, s) \iff \top, \quad \text{POSS}(\text{TURNRIGHT}, s) \iff \top.$$

Successor-state for Orientation. The orientation changes only on a turn:

$$\begin{aligned} \text{ORIENTATION}(\text{AGENT}, d, \text{RESULT}(a, s)) \iff & (a = \text{TURNLEFT} \wedge d = \text{LEFTTURN}(\text{ORI}(s))) \\ & \vee (a = \text{TURNRIGHT} \wedge d = \text{RIGHTTURN}(\text{ORI}(s))) \\ & \vee (a \neq \text{TURNLEFT} \wedge a \neq \text{TURNRIGHT} \\ & \wedge \text{ORIENTATION}(\text{AGENT}, d, s)) \end{aligned}$$

What examiners want you to demonstrate

- One **possibility axiom per action**, one **successor-state axiom per fluent** — not the other way round. A common error is to write effect axioms only and leave fluents un-framed.
- The successor-state axiom must close over *all* actions: the positive part lists actions that make F true, the frame part says F persists unless an action listed in the negative part undoes it.
- Quantify universally over a and s at the outermost level (omitted above for clarity, but should be made explicit in a formal answer): each axiom is $\forall a, s, \vec{x}. \dots$
- Re-use static predicates (WALL, ADJACENT, INLINEOFFIRE) and treat them as non-fluents — they need no successor-state axioms.
- Mention briefly that this formulation **solves the frame problem**: instead of $O(\text{actions} \times \text{fluents})$ frame axioms, you get one axiom per fluent.